

Bronchoscopy simulation: a brief review

Mohsen Davoudi · Henri G. Colt

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Abstract More than 500,000 flexible bronchoscopies are performed annually by chest physicians in the United States (Ernst et al., *Chest* 123:1693–1717, 2003). Indications include diagnosis of lung cancer and airway tumors, benign strictures, pulmonary infections, and treatment of central airway obstruction, emphysema, and intraepithelial lesions such as carcinoma in-situ. Anesthesiologists, cardiothoracic and trauma surgeons, otolaryngologists, and critical care physicians also perform this procedure as part of difficult airway management, intubation or airway inspection and bronchial toilet. Compared to the expanding body of simulation-related literature that is available in other procedural fields, however, the volume of published work relating to bronchoscopy is scant. The purpose of this paper is to provide an overview of the available literature pertaining to the use of simulation in bronchoscopy education, and to demonstrate how this limited yet valuable body of work lays a foundation for the future use of simulator-based bronchoscopy training.

Keywords Bronchoscopy · Education · Simulation · Competency

Procedural training in pulmonary fellowship programs has been inconsistent and variable (Ernst et al. 2003; Pastis et al. 2005; Haponik et al. 2000). Novice bronchoscopists learn to manipulate the flexible fiberoptic or video bronchoscope by watching faculty perform procedures, and then by performing examinations themselves under faculty supervision. In this apprenticeship model, technical, decision-making and interpretation skills are gradually acquired through patient-care experience. This method of learning, however, does not readily provide learners with consistent feedback, objective measures of skill acquisition,

M. Davoudi · H. G. Colt (✉)

Division of Pulmonary and Critical Care Medicine, University of California, Irvine, Medical Center,
101 City Drive South, Room 119, Bldg. 53, Rt. 81, Orange, CA 92868, USA
e-mail: hcolt@uci.edu

M. Davoudi
e-mail: mdavoudi@uci.edu

or opportunities to repeat and correct faulty maneuvers without placing the burden of procedure-related training on individual patients.

Despite the widespread practice of bronchoscopy, there are no firm guidelines that assure a uniform acquisition of basic or advanced skills necessary to achieve competency. Expected proficiency based on an arbitrary minimum numbers of procedures has been proposed (Ernst et al. 2003; Torrington 2000), yet the utility of such numeric criteria has been questioned (Alvarez et al. 2004; Prakash 2003). Considering the number of studies demonstrating the efficacy (Seymour et al. 2002), cost-effectiveness (Scott et al. 2000), and advantages of simulation training in terms of patient safety (Martin et al. 1998) and comfort (Sedlack et al. 2004), one would expect bronchoscopy educators to embrace this new learning modality.

The literature

A literature search crossing the words “bronchoscopy” and “simulation” on the National Library of Medicine’s Pub-Med returns 58 articles; the same search in the Ovid Gateway returns 149 publications. Of these, only 10 however are relevant to simulation in flexible bronchoscopy education (date accessed: August 1, 2007). The first published study by Ram et al. (1999) reports the use of an animal model (using swine airways) to simulate diagnostic and therapeutic bronchoscopy, including lavage, biopsy and the removal of foreign bodies for otolaryngology trainees. Also in 1999, Bro-Nielsen et al. (1999) published their first report of the development of an electronic hi-fidelity virtual reality (VR) simulator that could be used for bronchoscopy training; the “PreOp Endoscopic Simulator” (HT Medical Systems, Rockville, MD, USA) integrates force feedback, multimedia, and 3D graphics on a PC to give a relatively realistic bronchoscopy experience. The authors discuss the challenges, trade-offs and solutions developed in analyzing and prioritizing the tasks necessary to perform flexible bronchoscopy, and how simulation and multimedia technology were blended together to ensure adequate immersion and training efficacy. A follow-up study by Tasto et al. (2000) reports the modification of the same simulator for usage in flexible sigmoidoscopy and ureteroscopy.

The first report of the efficacy of simulation in bronchoscopy education (Colt et al. 2001) was published in 2001. In this prospective study, five novice bronchoscopists were taught to perform inspection flexible bronchoscopy using the same VR bronchoscopy simulator; training outcomes (dexterity, speed, and accuracy) were tested using the simulator and an inanimate airway model before and after 4 h of supervised training on the simulator combined with video-based training and 4 h of individual unsupervised practice. Novices significantly improved their dexterity and accuracy (missed fewer segments and had fewer contacts with the bronchial wall) in both models. When compared to a control group of four skilled physicians who had performed at least 200 bronchoscopies, the novice group equaled or surpassed the skilled physicians after the 8 h of training.

A study by Ost et al. (2001) validated the AccuTouch Flexible Bronchoscopy Simulator (Immersion Medical, Gaithersburg, MD, USA). In the initial observational phase of the study 11 novice, 8 intermediate, and 9 expert bronchoscopists were asked to perform a diagnostic bronchoscopy using the simulator; expected skill level was shown to be significantly correlated with the outcome measures (procedure time, time in red-out, number of scope collisions, percentage of segments entered, and use of suction). In the second randomized phase, two groups of three novice fellows were randomized to either training with the simulator or the traditional apprenticeship model. They were subsequently scored

for the outcome measures on two “solo” bronchoscopies on patients. The simulator group showed significant superiority in some but not all of the tested performance measures.

Rowe and Cohen (2002) demonstrated that even a single short session of training with a simulator (average time less than 40 min) could improve bronchoscopic endotracheal intubation skills in pediatric residents. Blum et al. (2004) also found that a 1-h session of simulator training improved the bronchoscopy skills of a group five first-year surgical residents compared to a similar group of their peers without the training; performance after the 1-h session was equivalent to a group of second and third year residents with prior bronchoscopy experience.

Moorthy et al. (2003) confirmed that bronchoscopy simulation, even without personal instruction or feedback, is an effective tool for the teaching and assessment of bronchoscopy. Nine novice participants each undertook 7–10 sessions of simulator training; at the end of each session they were allowed to see the computer-generated assessment sheet. Their performance was compared to a group of nine experienced bronchoscopists who had performed 200–1,000 bronchoscopies. In their first session, the novice group was significantly worse than the expert group in all four parameters measured by the simulator; these measures include percentage of segments visualized, number of wall collisions, procedure time, and economy of movement (percentage of segments divided by time). By the fifth session of training, there was no significant difference between the two groups. Moorthy’s group also demonstrated the face validity of the VR bronchoscopy simulator using a questionnaire assay given to the expert group.

In a study using simulation to assess bronchoscopy skill, Crawford and Colt (2004) scored the ability of 12 pulmonary fellows performing 5 standard maneuvers on the simulator on demand; this was done in the context of a bronchoscopy competency assessment, including a written test of bronchoscopic knowledge. The study demonstrated that bronchoscopic skill, as measured with the VR simulator does not correlate with years of training, number of reported bronchoscopies, or score on a written test of bronchoscopy knowledge. All 12 surveyed fellows felt that the simulation provided a realistic and accurate representation of the experience of bronchoscopy, and that training with the simulator, especially early in the learning process, would improve their procedural skills.

In a recent study from Taiwan, Chen et al. (2006) have validated a new, low-cost hi-fidelity VR bronchoscopy simulator (only \$25,000), which the developers call CBBS (computer-based bronchoscopy simulator). Comparing the performance of 10 novice bronchoscopists trained with the CBBS with 10 experts, the authors demonstrate that the simulator discriminates between novice and expert, using criteria such as procedure time, segments entered and identified, scope collisions, and need for verbal instruction. In a survey, experts are asked to assess the realism of the simulator experience; the investigators report scores of 7.9 out of 10 for spatial realism and 7.0 out of 10 for overall satisfaction. A satisfaction survey, comparing the 10 CBBS-trained novices with 10 conventionally-trained peers, demonstrates significantly higher levels of interest in and satisfaction from learning.

In order to systematically analyze the existing literature pertaining to the use of simulation in bronchoscopy education, the 10 studies were reviewed using a format inspired by the BEME study conducted by Issenberg et al. (2005). Each study was analyzed in terms of research design, expected learning outcome from intervention, target learner population, strength of findings, and documented improvement in measure outcome (Table 1). In order to view these results from within the larger context of research on simulation in general, we also examined each study in order to determine the evidence of favorable features and uses of medical simulation. These were broken down into 10 categories (Table 2). According to

Table 1 Summary characteristics of seven studies exploring the utility of simulation in bronchoscopy

Study	Research design	Target population	Expected learning outcome	Strength of findings	Documented improvement
Colt et al. (2001)	Prospective cohort	5 Novice pulm. fellows 4 Senior pulm. fellows	↓Wall contact ↓Red-out ↓Segments missed ↓Procedure time	High	Yes
Ost et al. (2001)	(A) Prospective cohort (B) Randomized control trial	(A) 9 Experts 8 Intermediates 11 Novices (B) 6 1st year pulm. fellows	↓Wall collision ↓Red-out ↑Segments visualized ↓Procedure time	High	Yes
Rowe and Cohen (2002)	Randomized control trial	20 Novice ped. residents: 12 Study 8 Controls	Improved fiberoptic intubation: ↓Wall collision ↓Idle time	Mod	Yes
Moorthy et al. (2003)	Prospective parallel cohort	9 Novices 9 Experienced bronchoscopists	↑% Segments visualized ↓Wall collision ↓Procedure time ↑Economy of performance	High	Yes

Table 1 continued

Study	Research design	Target population	Expected learning outcome	Strength of findings	Documented improvement
Blum et al. (2004)	Randomized control trial (with one extra non-randomized arm)	10 Novice surg. residents (5 study and 5 controls) 3 Experienced surg. residents (2nd & 3rd year)	↑Exam thoroughness ↓Redundant entries ↓Procedure time ↓Verbal/physical assists	High	Yes
Crawford and Colt (2004)	Cross-sectional observation	12 Pulm. fellows	N/A (assessment only)	High	Yes
Chen et al. (2006)	(A) Randomized control trial (B) Observational cohort	20 Novice bronchoscopists 10 Expert bronchoscopists	(A) ↑Learner satisfaction ↑Learner interest ↑Bronchoscopy knowledge (B) ↑Segments seen/identified ↓Redundant entries ↓Wall collisions ↓Procedure time ↑Pathology identified	High	Yes

Studies arranged in chronological order. Pulm. = Pulmonary; Ped. = Pediatrics; Surg. = Surgery

Table 2 Summary score and breakdown of seven studies exploring the utility of simulation in bronchoscopy for the presence or absence of 10 favorable features and uses

Study	Feedback	Repetitive practice	Curriculum integration	Range of difficulty levels	Adaptable learning strategies	Captures clinical variation	Controlled environment	Individualized learning	Defined outcomes or benchmarks	Validity of simulator	Study score
Colt et al. (2001)	✓	✓	✓	X	X	X	✓	✓	✓	✓	7/10
Ost et al. (2001)	✓	✓	X	X	X	X	✓	X	✓	✓	5/10
Rowe and Cohen (2002)	✓	✓	X	X	X	X	✓	X	✓	X	4/10
Moorthy et al. (2003)	✓	✓	X	X	X	X	✓	X	✓	✓	5/10
Blum et al. (2004)	X	✓	X	X	X	X	✓	X	✓	✓	4/10
Crawford and Colt (2004)	✓	X	✓	X	X	X	✓	X	✓	✓	5/10
Chen et al. (2006)	✓	✓	X	✓	X	✓	✓	✓	✓	✓	8/10

✓ = Feature or usage explored; X = Feature or usage not explored

Issenberg et al. (2005), the possibility to offer these 10 features and usages is what makes simulation preferable to conventional educational models; they are (1) feedback, (2) repetitive practice, (3) curriculum integration, (4) varying levels of difficulty, (5) multiple learning strategies, (6) capturing clinical variation, (7) controlled environment, (8) individualized learning, (9) defined outcomes or benchmarks, and (10) simulator validity (Table 2). We assessed each individual study to determine if it had explored the presence and utility of each of these 10 features in the simulation model being studied. We assigned a summary score out of 10 to each study, based on the total number of favorable features and uses explored.

One critical issue pertaining to the efficacy of any simulator-based procedural training is transfer: do the skills acquired in the simulated setting translate to real patient-care skills? Of the studies reviewed, the only one that attempts to address this issue is the study by Ost et al. (2001). In this study, however, the number of participants was limited to three in the simulator-trained and control groups, and the outcome measures employed were less than optimum: quantitative scoring [(% of segments visualized and correctly identified)/(total time in seconds)] was done by non-blinded faculty, and qualitative scoring was subjectively performed by a bronchoscopy nurse based on a composite value encompassing “the overall quality of the bronchoscopy in terms of patient preparation and anesthesia, patient comfort, speed, skill, coughing, and teamwork and communication with the bronchoscopy nursing staff”. The recent development and validation of objective instruments measuring bronchoscopy skills (Davoudi et al. 2006) could facilitate future studies attempting to tackle this vital issue.

Our findings suggest a relative deficit in research and analysis pertaining to bronchoscopy education. Of the 10 studies cited above, 3 are descriptive reports of animal or simulated models in which usage was not assessed. The 7 remaining studies were performed with few subjects and varying methodologies. Although demonstrating improvement in the expected learning outcomes, many of the desirable features were not explored; critical issues such as curriculum integration, varying levels of difficulty, adaptable learning strategies, capturing clinical variation or individualized learning were seldom or never addressed. This shortcoming is clearly demonstrated by the fact that the observed median summary score among the 7 studies was 5 out of 10 (Table 2).

The technology

The technology that is commercially available today for bronchoscopy simulation can be divided into two broad categories: hi-fidelity computer-based electronic simulation, and lo-fidelity mechanical inanimate airway models. The AccuTouch[®] bronchoscopy simulator is the product of the merger of HT Medical and Immersion Corporation (Gaithersburg, MD, USA). It consists of a proxy bronchoscope, a robotic interface device, and a personal computer with monitor and simulation software. The proxy bronchoscope, modeled after a conventional bronchoscope, is manually inserted into the simulator through the nostril of a plastic face model. The robotic interface device tracks the motions of the proxy, reproducing the forces felt during bronchoscopy, while the monitor displays computer-generated realistic images of the airway as the user navigates through the virtual anatomy. The virtual patient breathes, coughs, bleeds, and exhibits changes in vital signs. Different scenarios and diagnoses allow the learner to examine a normal airway, inspect airway tumors, and perform biopsies, brushing, bronchoalveolar lavage and transbronchial needle aspiration. As in real bronchoscopy, touching the walls of the virtual airway prompts cough;

a simulation of mucus on the tip of the scope can cause a white blur; a reddish blur (red out) is seen when the tip of the proxy bronchoscope is inadvertently wedged up against virtual airway mucosa. An optional roadmap feature can indicate the position of the proxy bronchoscope, while the simulator records data such as the duration of each procedure, the amount of topical anesthetic used, the number and sequence of bronchial segments inspected, the number of collisions with the tracheobronchial wall, and the duration of red out. The only other VR airway simulator that the authors are aware of is the Taiwanese CBBS, described in the study by Chen et al. From the brief description provided in the publication (Chen et al. 2006), the CBBS seems to be similar in many respects to the AccuTouch model.

The authors are aware of two available lo-fidelity mechanical inanimate airway models. The CLA Broncho Boy (CLA, Coburg, Germany) was developed in the late 1970s by Doctor John Nakhosteen, a professor of pulmonary medicine at the University of Bochum, Germany. This model consists of a head and torso, with complete upper airway, larynx, and tracheobronchial tree to the level of 1st sub-segmental bronchus, molded from a single piece of silicone. The flexible bronchoscope can be advanced through the mouth or nose; this model can be also used for learning rigid bronchoscopy, as well as for performing laryngo-tracheal intubation and airway management exercises. The Laerdal Airway Management Trainer (Laerdal, Stavanger, Norway) is a similar model, in which the trachea and left and right bronchial trees are made of separate pieces of silicone. These trainers vary in cost between \$1,500 and \$4,500. A group of Italian investigators recently proposed a far less expensive alternative. Di Domenico et al. (2007), use simple material similar to papier-mâché costing \$20 to create a real-scale and anatomically accurate tracheo-bronchial tree. In addition to the VR simulators and inanimate models, live and euthanized animals are still used for training in some institutions. In light of the existing technology and available models, we cannot ethically advocate or justify this practice.

Discussion

The studies reviewed, have established an adequate groundwork for the use of simulation in bronchoscopy education. It has been demonstrated that learning bronchoscopy using simulation improves variables such as procedure efficiency and economy of movement, thoroughness and accuracy of the airway examination, and decreases airway wall trauma, while enhancing learner satisfaction and interest. Current available hi-fidelity VR and inanimate models for simulation provide the prerequisite tools. Yet, even in industrial nations, only a small percentage of training programs are beginning to employ simulation as a mode of structured bronchoscopy training prior to patient contact.

One of the primary challenges facing bronchoscopy educators is the prohibitive cost of the available simulation tools. The common VR simulator used in the United States is now sold at over \$120,000. The paucity of simulation research companies designing airway scenarios seems to have limited competition that could have positively impacted the market. Innovations by groups like Chen et al. (2006) and Di Domenico et al. (2007) can be a stimulus for competition. We must help manufacturers recognize the existence of a potential worldwide market, especially if the prohibitive prices are broken.

In other procedural fields, investigators have explored the equivalence or superiority of hi- and lo-fidelity simulation models (Matsumoto et al. 2002; Grober et al. 2004). No such studies have yet been conducted in bronchoscopy. Such research, especially if it concludes the non-inferiority of inexpensive lo-fidelity mechanical inanimate airway models, will be

an important step in lowering the cost obstacle to more widespread usage of simulator-based bronchoscopy training.

Another challenge, which needs to be overcome, is the resistance of some bronchoscopy educators and pulmonary program directors, many of who remain attached to the old apprenticeship model. This perceived lack of necessity for simulation, in addition to the absence of a generally approved comprehensive bronchoscopy curriculum, has distinguished bronchoscopy from many of the other endoscopic and minimally invasive procedural fields. A move toward curriculum development (Haponik et al. 2000; Prakash 2003; Lund 2006), and increasing emphasis on competency-based training and patient safety will hopefully assist simulation advocates surmount these hurdles.

A third, and perhaps most important challenge to overcome is the prevailing perception among end-users of bronchoscopy education (members of national and international pulmonary and bronchology societies, along with trainees and specialists in practice) that bronchoscopy is minimally invasive, safe, and satisfactorily learned when patients bear the burden of procedure-related training (Ouellette 2006). Considering the increased number of indications for various diagnostic and therapeutic bronchoscopic procedures, the diverse medical and surgical specialties apt to perform these procedures, and in view of currently available computer-based and inanimate models, we submit this paradigm to be indefensible.

It is indeed fortunate, therefore, that recent evidence points to a change in these perceptions: the American College of Chest Physicians, for example, has introduced a simulation center that includes bronchoscopy simulation into its yearly international meeting. It is becoming increasingly mainstream to advocate that patients need no longer bear the burden of bronchoscopy-related training. Practitioners, as well as bronchoscopists in training, should be continuously exposed to the many advantages of simulation, emphasizing the immense potential to learn new skills as well as improve upon old ones, while protecting patients from undue harm and discomfort.

Additional systematic research is needed in order to explore issues such as the possibility of integration of bronchoscopy simulation in the framework of a bronchoscopy curriculum and assessment measures (Davoudi et al. 2006, 2007), adaptation of learning strategies to diverse settings and different subspecialty needs, the capture of clinical variation and varying levels of difficulty for a more realistic simulated experience, and methods to individualize the simulated learning experience. The results from large multi-center studies will need to ultimately demonstrate improved educational and patient-care outcomes, and most importantly explore the transfer of skills to the bedside. The prerequisite for such research is consensus-based competency-oriented curricular design: curricula, which combine the positive attributes of simulation, model-based practice, and the traditional apprenticeship mode of learning.

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